

Exercise 1C — Full Solutions

Questions 1–9 (complete)

Roots of quartic equations: power sums via the recurrence relation

Colour key: *green* = verification check | *blue* = alternative method.

For a quartic $ax^4 + bx^3 + cx^2 + dx + e = 0$ with roots $\alpha, \beta, \gamma, \delta$, Vieta gives

$$\sum \alpha = -\frac{b}{a}, \quad \sum \alpha\beta = \frac{c}{a}, \quad \sum \alpha\beta\gamma = -\frac{d}{a}, \quad \alpha\beta\gamma\delta = \frac{e}{a}.$$

Power-sum recurrence (multiply a root's equation $ar^4 + br^3 + cr^2 + dr + e = 0$ by r^n and sum over all four roots):

$$aS_{n+4} + bS_{n+3} + cS_{n+2} + dS_{n+1} + eS_n = 0, \quad S_0 = 4.$$

Seeds: $S_1 = \sum \alpha$ and $S_{-1} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta}$ from Vieta; $S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta$ by squaring the sum (this avoids the S_{-2} term that the recurrence would need at $n = -2$). The recurrence then delivers S_4 (at $n = 0$) and S_3 (at $n = -1$).

Question 1

For each quartic, find $\sum \alpha$ and $\sum \alpha\beta$, i.e. read off $-\frac{b}{a}$ and $\frac{c}{a}$.

(a) $x^4 - 2x^3 + 5x^2 + 7 = 0$ ($a = 1, b = -2, c = 5$): $\sum \alpha = -\frac{-2}{1} = 2, \quad \sum \alpha\beta = \frac{5}{1} = 5.$

(b) $2x^4 + 5x^3 - 3x + 4 = 0$ ($a = 2, b = 5, c = 0$): $\sum \alpha = -\frac{5}{2}, \quad \sum \alpha\beta = \frac{0}{2} = 0.$

(c) $3x^4 - 2x^2 + 9x - 11 = 0$ ($a = 3, b = 0, c = -2$): $\sum \alpha = -\frac{0}{3} = 0, \quad \sum \alpha\beta = -\frac{2}{3}.$

Check. Where a coefficient is absent the corresponding symmetric function vanishes: (b) has no x^2 term so $\sum \alpha\beta = 0$ ✓; (c) has no x^3 term so $\sum \alpha = 0$ ✓.

Question 2

$5x^4 - 3x^3 + x - 13 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find (a) $\sum \alpha$ and $\sum \alpha^2$; (b) $\sum \frac{1}{\alpha}$.

Here $a = 5, b = -3, c = 0, d = 1, e = -13$, so by Vieta

$$\sum \alpha = -\frac{b}{a} = \frac{3}{5}, \quad \sum \alpha\beta = \frac{c}{a} = 0, \quad \sum \alpha\beta\gamma = -\frac{d}{a} = -\frac{1}{5}, \quad \alpha\beta\gamma\delta = \frac{e}{a} = -\frac{13}{5}.$$

(a) $\sum \alpha = \frac{3}{5}$, and $\sum \alpha^2 = (\sum \alpha)^2 - 2\sum \alpha\beta = \left(\frac{3}{5}\right)^2 - 0 = \frac{9}{25}.$

(b) $\sum \frac{1}{\alpha} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{-1/5}{-13/5} = \frac{1}{13}.$

Check. Numerically $\sum \alpha = 0.600$, $\sum \alpha^2 = 0.360 = \frac{9}{25}$, $\sum \frac{1}{\alpha} = 0.0769 = \frac{1}{13}$ ✓.

Question 3

$x^4 + x + 2 = 0$ has roots $\alpha, \beta, \gamma, \delta$. State S_1 and S_{-1} , and find S_2 .

Here $a = 1, b = 0, c = 0, d = 1, e = 2$, so by Vieta

$$\sum \alpha = 0, \quad \sum \alpha\beta = 0, \quad \sum \alpha\beta\gamma = -\frac{d}{a} = -1, \quad \alpha\beta\gamma\delta = \frac{e}{a} = 2.$$

S_1 : $S_1 = \sum \alpha = 0$ (no x^3 term).

$$S_{-1}: S_{-1} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{-1}{2} = -\frac{1}{2}.$$

S_2 : squaring the sum, $S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta = 0 - 0 = 0$.

Check. $S_2 = 0$ with four roots not all zero forces non-real roots. Numerically the quartic has two complex conjugate pairs, and $S_2 = 0.000$, $S_{-1} = -0.500$ ✓.

Question 4

$2x^4 + x^3 - x + 7 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Given $S_3 = \frac{11}{8}$, use the recurrence to find S_4 .

Here $a = 2, b = 1, c = 0, d = -1, e = 7$, so the recurrence (multiply a root's equation by y^n and sum) is

$$2S_{n+4} + S_{n+3} + 0 \cdot S_{n+2} - S_{n+1} + 7S_n = 0, \quad S_0 = 4.$$

We also need $S_1 = \sum \alpha = -\frac{b}{a} = -\frac{1}{2}$. Apply the recurrence at $n = 0$ (leading term $2S_4$):

$$2S_4 + S_3 - S_1 + 7S_0 = 0 \implies 2S_4 + \frac{11}{8} - \left(-\frac{1}{2}\right) + 7(4) = 0 \implies 2S_4 + \frac{239}{8} = 0 \implies S_4 = -\frac{239}{16}.$$

Check. Numerically $S_3 = 1.375 = \frac{11}{8}$ (as given) and $S_4 = -14.9375 = -\frac{239}{16}$ ✓.

Question 5

$x^4 - x^3 + x + 2 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find $\sum \alpha$, $\sum \alpha^2$, $\sum \frac{1}{\alpha}$, and hence $\sum \alpha^3$.

Here $a = 1, b = -1, c = 0, d = 1, e = 2$, so by Vieta

$$\sum \alpha = 1, \quad \sum \alpha\beta = 0, \quad \sum \alpha\beta\gamma = -\frac{d}{a} = -1, \quad \alpha\beta\gamma\delta = \frac{e}{a} = 2.$$

$$\sum \alpha = S_1 = 1, \quad \sum \alpha^2 = S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta = 1 - 0 = 1, \quad \sum \frac{1}{\alpha} = S_{-1} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta} = -\frac{1}{2}.$$

$\sum \alpha^3 = S_3$ **by the recurrence at $n = -1$:** $aS_3 + bS_2 + cS_1 + dS_0 + eS_{-1} = 0$ with $S_0 = 4$:

$$S_3 - (1) + 0 + (1)(4) + (2)\left(-\frac{1}{2}\right) = 0 \implies S_3 - 1 + 4 - 1 = 0 \implies S_3 = -2.$$

Alternative for S_3 — factorisation identity. $\sum \alpha^3 = 3\sum \alpha\beta\gamma + (\sum \alpha)(\sum \alpha^2 - \sum \alpha\beta) = 3(-1) + (1)(1 - 0) = -3 + 1 = -2$.

Check. Numerically $\sum \alpha = 1.000$, $\sum \alpha^2 = 1.000$, $\sum \frac{1}{\alpha} = -0.500$, $\sum \alpha^3 = -2.000$ ✓.

Question 6

$x^4 + ax^2 + bx + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Given $S_2 = S_{-1}$, find S_3 in terms of a .

The coefficients are (leading 1, then $0, a, b, 1$), so by Vieta

$$\sum \alpha = 0, \quad \sum \alpha\beta = a, \quad \sum \alpha\beta\gamma = -b, \quad \alpha\beta\gamma\delta = 1.$$

Then $S_1 = \sum \alpha = 0$, $S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta = -2a$, and $S_{-1} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta} = -b$. The condition $S_2 = S_{-1}$ gives

$$-2a = -b \implies b = 2a.$$

S_3 by the recurrence at $n = -1$: $S_{n+4} + 0 \cdot S_{n+3} + aS_{n+2} + bS_{n+1} + S_n = 0$ gives, with $S_0 = 4$,

$$S_3 + aS_1 + bS_0 + S_{-1} = 0 \implies S_3 + 0 + 4b - b = 0 \implies S_3 = -3b = -6a.$$

Alternative for S_3 — factorisation identity. $\sum \alpha^3 = (\sum \alpha)^3 - 3(\sum \alpha)(\sum \alpha\beta) + 3\sum \alpha\beta\gamma = 0 - 0 + 3(-b) = -3b = -6a.$

Check. Take $a = 3$ (so $b = 6$): the quartic $x^4 + 3x^2 + 6x + 1 = 0$ gives $S_2 = -6 = S_{-1}$ ✓ and $S_3 = -18 = -6a$ ✓.

Question 7

$3x^4 + 2x^3 + 7x^2 + 4 = 0$, with $S_n = \sum \alpha^n$. (a) find S_1, S_2 ; (b) find S_3, S_4 ; (c) any complex roots?

Here $a = 3, b = 2, c = 7, d = 0, e = 4$, so by Vieta

$$\sum \alpha = -\frac{2}{3}, \quad \sum \alpha\beta = \frac{7}{3}, \quad \sum \alpha\beta\gamma = -\frac{d}{a} = 0, \quad \alpha\beta\gamma\delta = \frac{e}{a} = \frac{4}{3}.$$

(a) $S_1 = \sum \alpha = -\frac{2}{3}, \quad S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta = \frac{4}{9} - \frac{14}{3} = -\frac{38}{9}.$

(b) First $S_{-1} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{0}{4/3} = 0.$

Recurrence at $n = -1$: $3S_3 + 2S_2 + 7S_1 + 0 \cdot S_0 + 4S_{-1} = 0$:

$$3S_3 + 2(-\frac{38}{9}) + 7(-\frac{2}{3}) + 0 = 0 \implies 3S_3 = \frac{76}{9} + \frac{42}{9} = \frac{118}{9} \implies S_3 = \frac{118}{27}.$$

Recurrence at $n = 0$: $3S_4 + 2S_3 + 7S_2 + 0 \cdot S_1 + 4S_0 = 0$ with $S_0 = 4$:

$$3S_4 + 2(\frac{118}{27}) + 7(-\frac{38}{9}) + 16 = 0 \implies 3S_4 + \frac{236}{27} - \frac{798}{27} + \frac{432}{27} = 0 \implies S_4 = \frac{130}{81}.$$

(c) $S_2 = -\frac{38}{9} < 0$. A sum of squares of *real* numbers cannot be negative, so the roots are **not all real** — there is at least one complex conjugate pair.

Alternative for S_3 — factorisation identity. $\sum \alpha^3 = 3\sum \alpha\beta\gamma + (\sum \alpha)(\sum \alpha^2 - \sum \alpha\beta) = 3(0) + (-\frac{2}{3})(-\frac{38}{9} - \frac{7}{3}) = (-\frac{2}{3})(-\frac{59}{9}) = \frac{118}{27}.$

Check. Numerically $S_1 = -0.667, S_2 = -4.222, S_3 = 4.370, S_4 = 1.605$, agreeing with the exact values found above ✓. All four roots are complex (two conjugate pairs), consistent with (c).

Question 8

$x^4 + ax^3 + bx^2 + c = 0$ has roots $\alpha, \beta, \gamma, \delta$. Given $\sum \alpha = 2, \alpha\beta\gamma\delta = 1$ and $\sum \alpha^2 = 0$, find a, b, c .

The coefficients are (leading 1, then $a, b, 0, c$ — note *no* x term), so by Vieta

$$\sum \alpha = -a, \quad \sum \alpha\beta = b, \quad \alpha\beta\gamma\delta = c.$$

a : $\sum \alpha = -a = 2 \Rightarrow a = -2.$ c : $\alpha\beta\gamma\delta = c = 1 \Rightarrow c = 1.$

b : $\sum \alpha^2 = (\sum \alpha)^2 - 2\sum \alpha\beta = 2^2 - 2b = 0 \Rightarrow b = 2.$

Hence $\boxed{a = -2, b = 2, c = 1}.$

Check. The quartic is $x^4 - 2x^3 + 2x^2 + 1 = 0$. Numerically $\sum \alpha = 2.000$, $\alpha\beta\gamma\delta = 1.000$, $\sum \alpha^2 = 0.000$ ✓.

Question 9

The quartic $x^4 - 2x^3 + x^2 - 4 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Show that $S_4 = 9S_3$.

Here $a = 1, b = -2, c = 1, d = 0, e = -4$, so by Vieta

$$\sum \alpha = 2, \quad \sum \alpha\beta = 1, \quad \sum \alpha\beta\gamma = -\frac{d}{a} = 0, \quad \alpha\beta\gamma\delta = \frac{e}{a} = -4.$$

Seeds: $S_1 = \sum \alpha = 2$, $S_2 = (\sum \alpha)^2 - 2\sum \alpha\beta = 4 - 2 = 2$, $S_{-1} = \frac{0}{-4} = 0$.

Recurrence at $n = -1$: $S_3 - 2S_2 + S_1 + 0 \cdot S_0 - 4S_{-1} = 0$:

$$S_3 - 2(2) + 2 + 0 = 0 \implies S_3 = 2.$$

Recurrence at $n = 0$: $S_4 - 2S_3 + S_2 + 0 \cdot S_1 - 4S_0 = 0$ with $S_0 = 4$:

$$S_4 - 2(2) + 2 - 16 = 0 \implies S_4 = 4 - 2 + 16 = 18.$$

Therefore $S_4 = 18 = 9 \times 2 = 9S_3$, as required. ■

Alternative for S_3 — factorisation identity. $\sum \alpha^3 = 3\sum \alpha\beta\gamma + (\sum \alpha)(\sum \alpha^2 - \sum \alpha\beta) = 3(0) + (2)(2 - 1) = 2$, confirming $S_3 = 2$.

Check. Numerically $S_3 = 2.000$ and $S_4 = 18.000$, so $S_4/S_3 = 9$ exactly ✓.